TWO RELATIONS IN THE LOCAL WEYL MODULE ATTACHED TO A MULTIPLE OF THE HIGHEST ROOT

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Allowed standard prerequisites (editorial).

- The local Weyl module W((k+1)theta) with its highest weight vector, and the relation (w2) remembering that a sufficiently high power of a negative root vector acts as zero on it. The author's section 3, where the module and the relation are set up before the counted statement; the relation is referred to in the proof at source line 367 (a navigation-pointer stub). Fixes the module in which the two relations hold and supplies the vanishing used at the end of both parts.
- The commutator of a raising operator with a power of a lowering operator in the current algebra, and the evaluation of the coefficient appearing in the expansion. Computed inside the counted proof at source lines 341-356. Produces the explicit multiple of the desired vector which equals a vector known to vanish.
- The commutation relation for two negative root vectors, labelled garland in the article. Invoked inside the counted proof at source line 365 (a navigation-pointer stub). Gives the second relation after specialising the parameters.

Source reference. Original reference garland: sources/sigma-2014-110-demazure-fusion/pinned-original-source.tex, source lines 315–315. Navigation pointer only: the counted text refers to the item garland of this same article, which is not part of this card. The pointer carries the item's own author label as its number; its content is not reproduced here.

Source reference. Original reference w2: sources/sigma-2014-110-demazure-fusion/pinned-original-source.tex, source lines 307–307. Navigation pointer only: the counted text

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refers to the item w2 of this same article, which is not part of this card. The pointer carries the item's own author label as its number; its content is not reproduced here.

Lemma 1. *Let $k \in \mathbb{N}$. The following relations hold in the local Weyl module $W((k+1)\theta)$:*

1) $(x_\theta^- \otimes 1)^{2k+1}\big(x_\theta^- \otimes t^{2k+1-i}\big)w_{(k+1)\theta} = 0, \qquad \forall\, 0 \le i \le k;$
2) $(x_\theta^- \otimes t^m)(x_\theta^- \otimes t^{m+1})w_{(k+1)\theta} \in \big\langle \big(x_\theta^- \otimes t^{m+2}\big)w_{(k+1)\theta}\big\rangle, \qquad \forall\, m \ge k.$

Proof. To prove part (1), consider $(x_\theta^+ \otimes t^{2k+1-i})(x_\theta^- \otimes 1)^{2k+3}w_{(k+1)\theta}$. Since $(x_\theta^+ \otimes t^{2k+1-i})w_{(k+1)\theta} = 0$, we get

$$\begin{aligned}\big(x_\theta^+ \otimes t^{2k+1-i}\big)(x_\theta^- \otimes 1)^{2k+3}w_{(k+1)\theta} &= \big[x_\theta^+ \otimes t^{2k+1-i}, (x_\theta^- \otimes 1)^{2k+3}\big]w_{(k+1)\theta} \\ &= \sum_{j=1}^{2k+3}(x_\theta^- \otimes 1)^{j-1}\big(\theta^\vee \otimes t^{2k+1-i}\big)(x_\theta^- \otimes 1)^{2k+3-j}w_{(k+1)\theta}.\end{aligned}$$

Since $(\theta^\vee \otimes t^{2k+1-i})w_{(k+1)\theta} = 0$, we may replace $(\theta^\vee \otimes t^{2k+1-i})(x_\theta^- \otimes 1)^{2k+3-j}$ by $\big[\theta^\vee \otimes t^{2k+1-i}, (x_\theta^- \otimes 1)^{2k+3-j}\big] = (-2)(2k+3-j)(x_\theta^- \otimes 1)^{2k+2-j}\big(x_\theta^- \otimes t^{2k+1-i}\big)$.

After simplifying, we get

$$\begin{aligned}&\big(x_\theta^+ \otimes t^{2k+1-i}\big)(x_\theta^- \otimes 1)^{2k+3}w_{(k+1)\theta} \\ &= (-1)(2k+2)(2k+3)(x_\theta^- \otimes 1)^{2k+1}\big(x_\theta^- \otimes t^{2k+1-i}\big)w_{(k+1)\theta}.\end{aligned}$$

Now, using $(x_\theta^- \otimes 1)^{2k+3}w_{(k+1)\theta} = 0$ in $W((k+1)\theta)$, completes the proof of part (1). Part (2) follows easily by putting $r = 2$, $s = 2m+1$ and $\alpha = \theta$ in Lemma garland, and using the fact that $(x_\theta^- \otimes 1)^{2m+3}w_{(k+1)\theta} = 0$, $\forall\, m \ge k$ by (w2). □